

MAT 310, Handout 16: Beta Distribution (4.7)

Beta Distribution $Y \sim \text{Beta}(\alpha, \beta)$ is Beta distributed with $\alpha > 0$ and $\beta > 0$ if its density function is given by

$$f(y) =$$

where the beta function is given by $B(\alpha, \beta) =$

Remark: Because $0 \leq y \leq 1$ for $Y \sim \text{Beta}(\alpha, \beta)$, the Beta distribution is often used to model proportions and probabilities.

Expected Value and Variance of $Y \sim \text{Beta}(\alpha, \beta)$

$$E(Y) =$$

$$V(Y) =$$

Example 1. The fraction of small businesses that show an annual profit in a given year is a random variable whose probability density function is given by $f(y) = \begin{cases} ky^2(1-y), & 0 \leq y \leq 1 \\ 0, & o.w. \end{cases}$.

- (a) What is the distribution of Y?
- (b) Find k .
- (c) What is the expected fraction of small businesses that show profit in a given year?
- (d) Find the probability that less than 50% of small businesses show profit in a given year.